D7043E · GROUP 17

Edge-Ready ECG Classification

Finding abnormal heartbeats on a microcontroller

CRISP-DM, end to end — 44 patients, 100,622 beats, four AAMI classes, one 35 KB CNN synthesised for the MAX78002.

100,622
beats prepared

35,588
parameters (1.5 % of memory)

11 / 11
toolchain checks passed

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Raw recordings to a model-ready dataset

MIT-BIH
48 records
2 leads, 360 Hz

›

Causal filter
0.5–40 Hz, 60 Hz notch
no future samples

›

Downsample
360 → 180 Hz
halves data memory

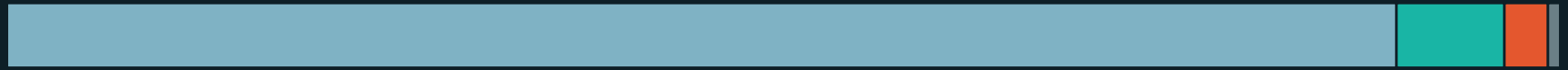
›

Normalise
per patient, robust
99.5th percentile

›

Segment
256-sample windows
1.42 s per beat

What came out — and why one class is hard



N 90,034 V 7,007 S 2,779 F 802 — the two rare classes are 3.6 % of the data

Split by patient, never by beat.
22 patients to train on, 22 held back for test — no one appears on both sides. A random beat split puts the same patient on both, and accuracy jumps to ~99 % for the wrong reason.

100,622
beats prepared

0
unexplained

Logistic Regression - Baseline

ECG Input

One lead at 180 Hz
256 samples per beat

DS1 Split

40,223 training beats
10,743 validation beats

Training

Training-set scaling
20:1 class weight cap

Model selection

C = 0.01, 0.1, 1 and 10
Validation macro-F1 for N/S/V

C = 10 Selected setting

0.4781 validation macro-F1 for N/S/V

82.19 % Validation accuracy

Noise Lowered Baseline accuracy on DS2

Metric	Clean DS2	0 dB SNR
Accuracy	**74.28%**	**64.50%**
Macro-F1 (N/S/V)	0.4433	0.3921
Macro-F1 (all four classes)	0.3429	0.3031

Same model, same beats

49,656 test beats
22 DS2 records

9.78

Percentage point drop in accuracy at 0 dB

Electrode-motion Noise was added to the raw ECG before pre-processing

A CNN shaped by the chip, not by taste



bar width = how much of the 1.42 s window is left after pooling

Stride fixed at 1

so pooling does every downsample

Pooling before convolution

the order the toolchain requires

ReLU only, no batch-norm

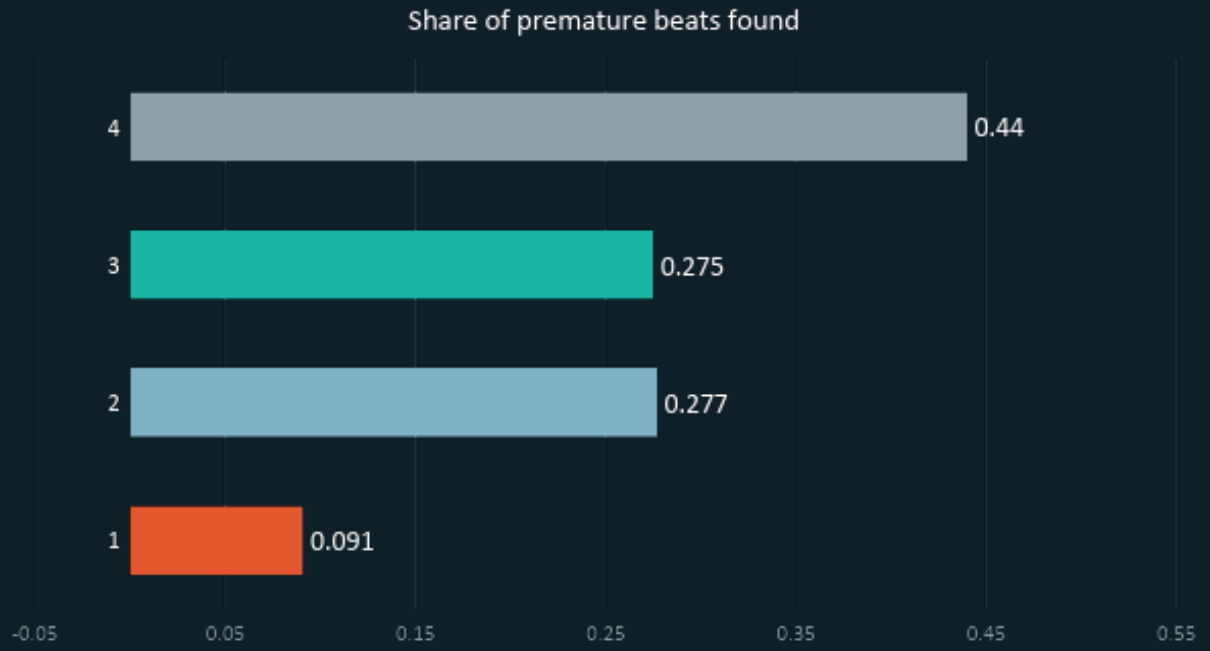
nothing else is supported

Classifier input ≤ 1024

the limit that decided our model

35,588 parameters **35 KiB** of the 2,340 KiB weight budget **1.5 %** used

One heartbeat cannot show a premature beat



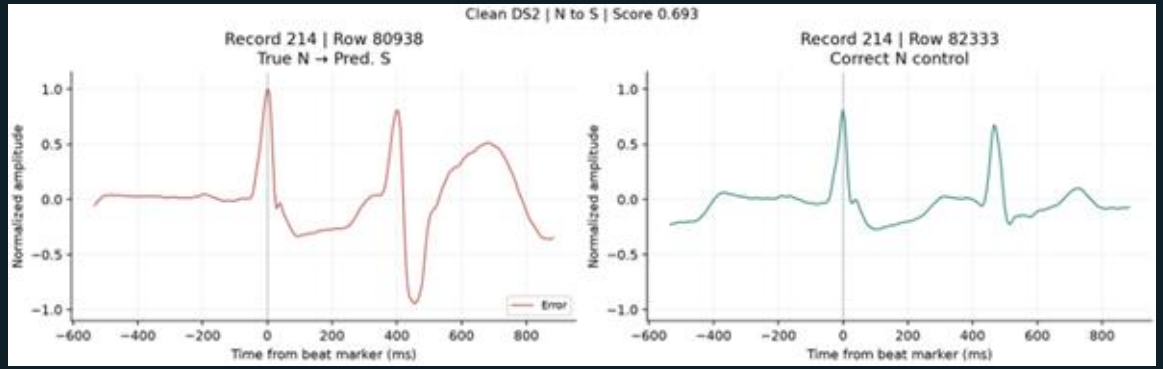
Why

An early beat is early. Timing lives between beats — one beat alone cannot carry it.

The catch

The longest bar false-alarms on 539 normal beats a run. We did not pick it.

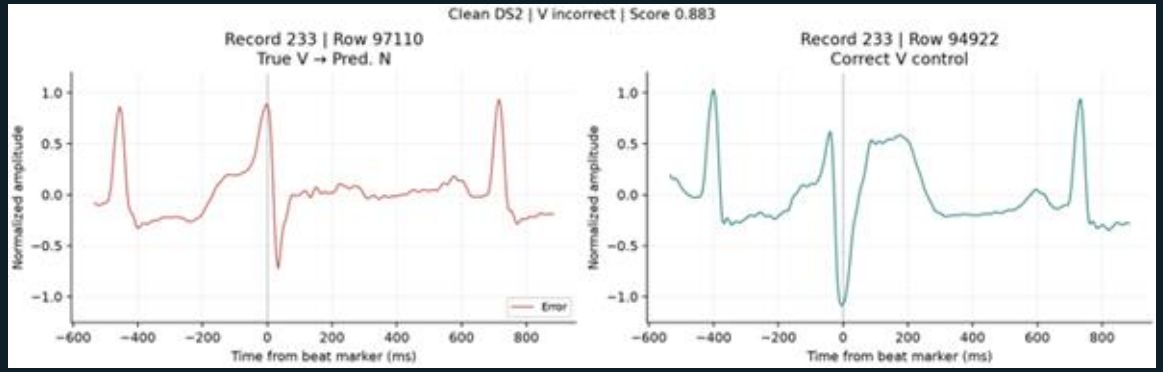
What the Baseline got wrong



N classified as S

Record 214, row 80938

The following beat has a deeper dip than in the correctly classified N example



V classified as N

Record 233, row 97110

The correctly classified V beat has a deeper dip near the beat marker

Each plot shows the error on the left and correctly classified beat of same class on the right

Test results on new patients (DS2)

Overall result

0.47

macro-F1 (baseline: 0.34)

Accuracy is 0.87. The FP32 and INT8 models give almost the same results.

Recall per class

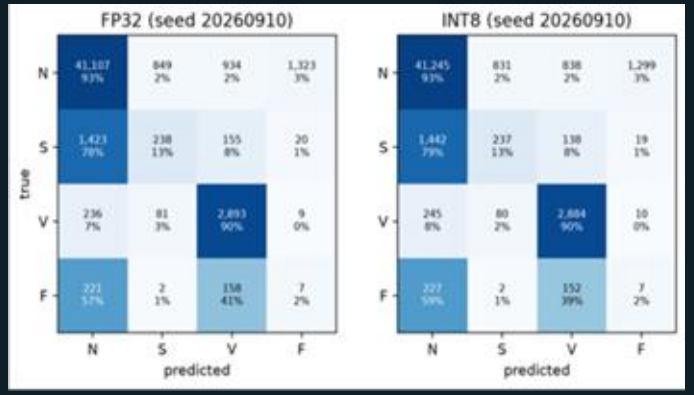
0.16

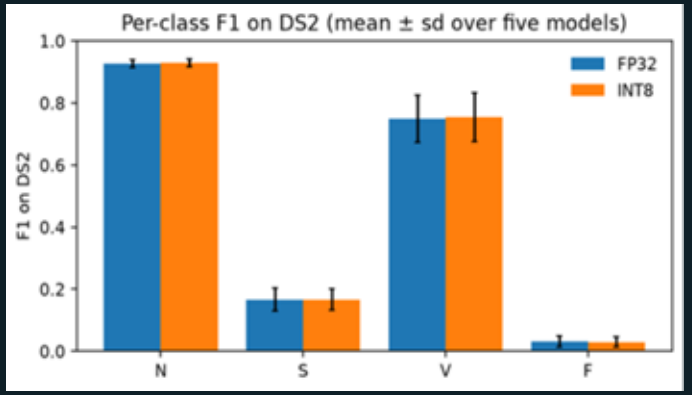
S recall (target: 0.60)

N 0.90, V 0.88, F 0.10. N and V beats are found well, S and F are not.

Record 232 has 75 % of all S beats in DS2. **The model finds only 2.9 % of them.** Without this patient, S recall is 0.54.

Where the model makes mistakes





Model S beats are predicted as N, and F beats are split between N and V. FP32 and INT8 make the same mistakes.

Did we reach our goals?

Achieved

- **Noise:** keeps 99 % at 12 dB and 97 % at 6 dB
- **Quantization:** the INT8 model is as good as FP32
- **Size:** 35,588 parameters, only 35 KB
- **MAX78002:** all 11 checks passed (software only)

Partially achieved

- **Macro-F1:** 0.47 is better than the baseline (0.34), but below our target of 0.55
- **Recall:** V reaches 0.88 (target 0.80), but S is only 0.16 (target 0.60)

4 achieved, 2 partially achieved. **The main reason is that very few patients have S beats,** not the size of the model.

From a trained model to C code for the chip

Re-express
ai8x layer classes
same 35,588 params

›

Train (QAT)
8-bit from epoch 60
best epoch 83

›

Quantize
int8 weights, biases
quantize.py

›

Simulate
8-bit arithmetic
validation data only

›

Synthesize
C project generated
ai8xize.py

Why the model fits the accelerator

Supported ops only
conv, max pool, ReLU, one FC layer

6 of 128 layers
pooling is fused into the convolutions

Final pool is its own layer
the chip cannot pool and flatten at once

Input 1 × 256, 8-bit
output: 4 logits, the largest is the class

35,588 parameters **6 / 128** hardware layers **1.5 %** of weight memory

The toolchain accepts it, and all 11 checks pass

Verification checks

11 / 11

independent checks passed

Weights are int8, and the synthesiser's simulation matches PyTorch's 8-bit logits bit for bit on 40 beats.

Resources on the chip

1.5 %

of weight memory — 35,408 of 2,396,160 bytes

Biases 2.2 %, activations about 8 KiB of 1,280 KiB, 1,660,928 MACs per beat.

Quantised model on validation: accuracy 0.944, macro-F1 (N/S/V) 0.681. S recall 0.24, F recall 0.00. The rare classes are still the weak spot.

Verified in software, not measured on a board

Verified in software

Toolchain accepts, quantises and synthesises

Memory use and 1.66 M MACs per beat

8-bit accuracy in simulation

Simulation bit-exact with PyTorch

Needs a MAX78002 board

Inference latency

Power and energy per inference

Accuracy on the real chip

Real-time streaming behaviour

No board, no hardware claim. **Nothing here was measured on a MAX78002.** Next: run the built-in known-answer test, compare logits beat by beat, then time and power.

The best model was not the best edge model

Highest score

1024 / 1024

classifier inputs used — no headroom

Quantisation and synthesis still ahead, with no room to give.

What we selected

512 / 1024

same score, half the width

Difference smaller than seed noise ($p = 0.46$). The metric could not choose — the hardware limit did.

We also miss our own target: **0.60 promised, 0.16 measured**. Only four patients in our training data have premature beats. Reported, not revised.

Testing the model's robustness to noise

Three types of noise

BASELINE WANDER

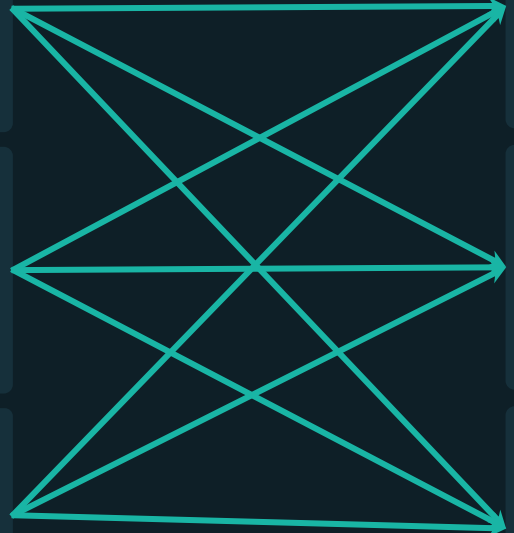
Low-frequency baseline shifts caused primarily by respiration or slow body movements.

ELECTRODE MOTION

Disturbances caused by the shifting or pulling of electrodes against the skin.

MUSCLE ARTIFACT

Higher-frequency electrical activity generated by skeletal muscle contractions.



Three levels of noise

12 dB

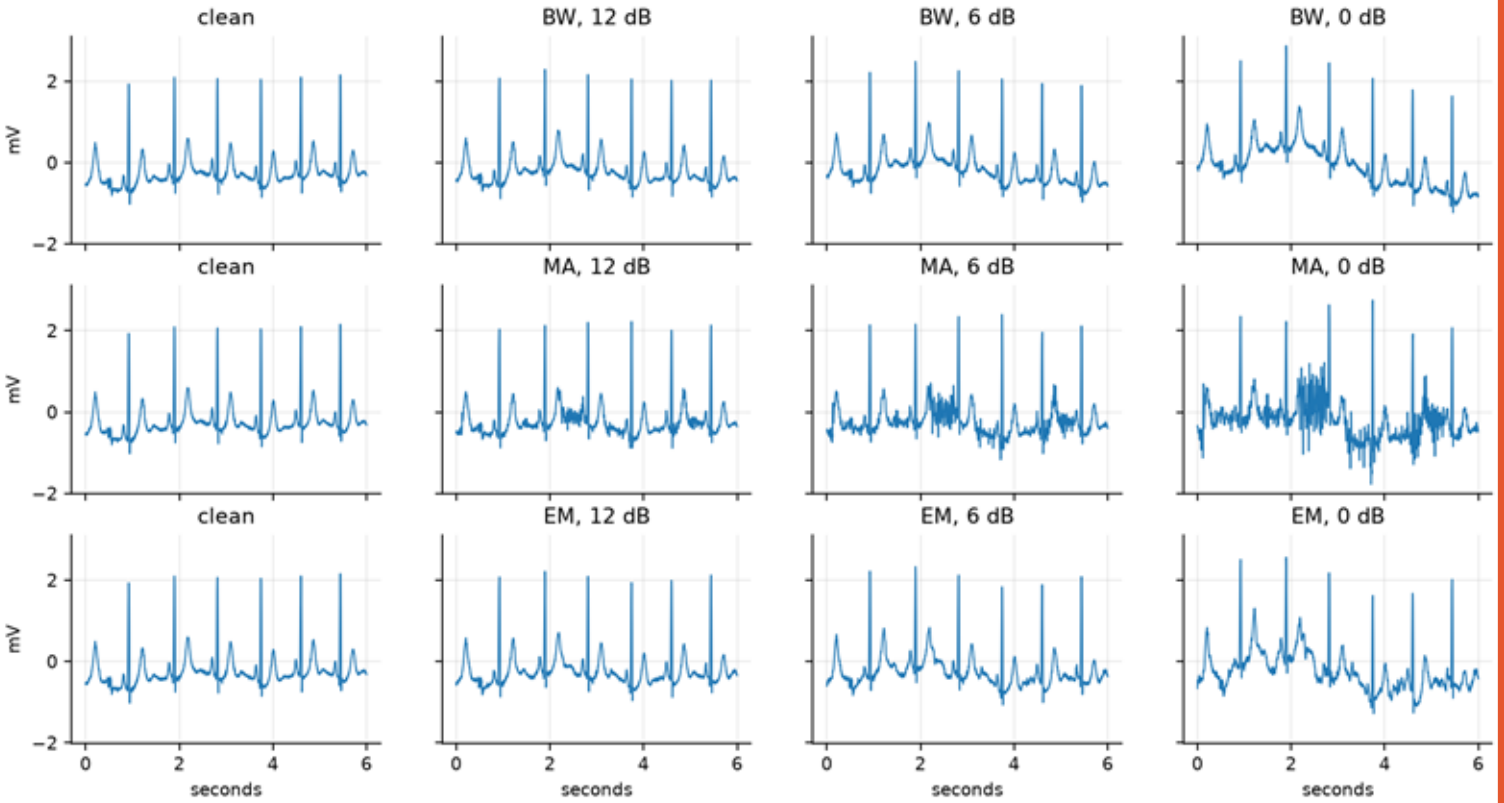
6dB

0dB

MIT-BIH Noise Stress Test Database

Levels of noise

Type of noise



GROUP 17 · D7043E · EDGE-READY ECG CLASSIFICATION

Thank you

Code, results and the generated MAX78002 C files:
github.com/ThaniJames/Group17_ECG_Project

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